

Polishetty Bala Arun

AI & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Result-oriented AI/ML Engineering student with hands-on experience in training deep learning models, building Retrieval-Augmented Generation (RAG) pipelines, and developing end-to-end web platforms. Proven capability in translating complex neural architectures into production-ready applications. Seeking to leverage skills in TensorFlow, LangChain, and Python to contribute to innovative AI initiatives.

EDUCATION

Vellore Institute of Technology (VIT), AP

Sept 2023 - Sept 2027

B.Tech. in Computer Science and Engineering with Specialization in AI & ML

GPA: 8.33 / 10.00

Little Flower Junior College, Hyderabad

Sept 2021 - April 2023

Intermediate Education (MPC - Mathematics, Physics, Chemistry)

Percentage: 92.9%

TECHNICAL SKILLS

Programming: Python, Java, SQL, C, C++

AI & Machine Learning: Deep Learning, Generative AI, RAG, LLMs (Zephyr-7B, OpenAI APIs), Natural Language Processing (NLP), Computer Vision, Explainable AI (XAI)

Frameworks & Libraries: TensorFlow, Keras, PyTorch, LangChain, Streamlit, Flask, React, Scikit-learn, Pandas, NumPy, OpenCV, FAISS, Sentence Transformers

Databases & Cloud: MySQL, PostgreSQL, SQLite, MongoDB, AWS (EC2, S3)

Developer Tools & Platforms: Git, GitHub Actions, Docker, VS Code, Jupyter Notebook, PyCharm, IntelliJ IDEA, Power BI, Tableau, Excel

PROJECTS

AI-Based Skin Disease Detection System

TensorFlow, Keras, OpenCV, Flask, React, NumPy

- Engineered an end-to-end AI dermatology assistant using deep convolutional neural networks (CNNs) in TensorFlow to classify skin diseases from medical images.
- Developed robust image pre-processing and data augmentation pipelines with OpenCV and NumPy, enhancing model generalization and reducing inference latency.
- Designed and deployed a responsive web interface using React and Flask to facilitate secure image uploads, asynchronous model predictions, and interactive visualization of results.

AI-Powered Drug Discovery Platform (DTI Prediction)

PyTorch, TensorFlow, React, Python, Plotly, Flask

- Architected a deep learning-based Drug-Target Interaction (DTI) prediction platform to accelerate virtual screening of candidate molecules from protein sequence data.
- Trained and evaluated CNN models in PyTorch to map chemical structures and protein sequences, predicting binding affinities with high precision.
- Integrated explainable AI (XAI) features using attribution maps and interactive Plotly dashboards, allowing researchers to visualize model decision boundaries.

Multi-PDF Retrieval-Augmented Generation (RAG) Application

LangChain, FAISS, Hugging Face Zephyr-7B, Streamlit, Sentence Transformers, PyPDFLoader

- Developed a high-performance RAG pipeline enabling context-aware question answering across multiple large PDF documents in real-time.
- Implemented document parsing and chunking using PyPDFLoader and Recursive Character Text Splitting, indexing text vector embeddings in a FAISS vector store.
- Integrated Zephyr-7B LLM via LangChain and designed a Streamlit interface, ensuring source-cited, hallucination-free responses based on retrieved context.

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional (Oracle)
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate (Oracle)
- AWS Academy Graduate - Cloud Architecting (Amazon Web Services)
- AWS Academy Graduate - Cloud Foundations (Amazon Web Services)
- SQL and Relational Databases 101 (IBM Skills Network)